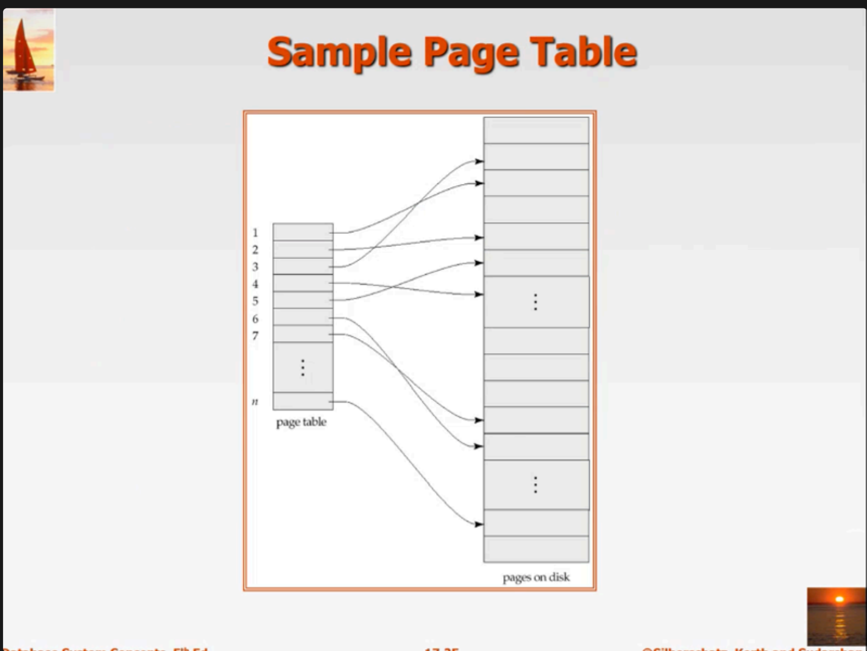
Shadow Paging

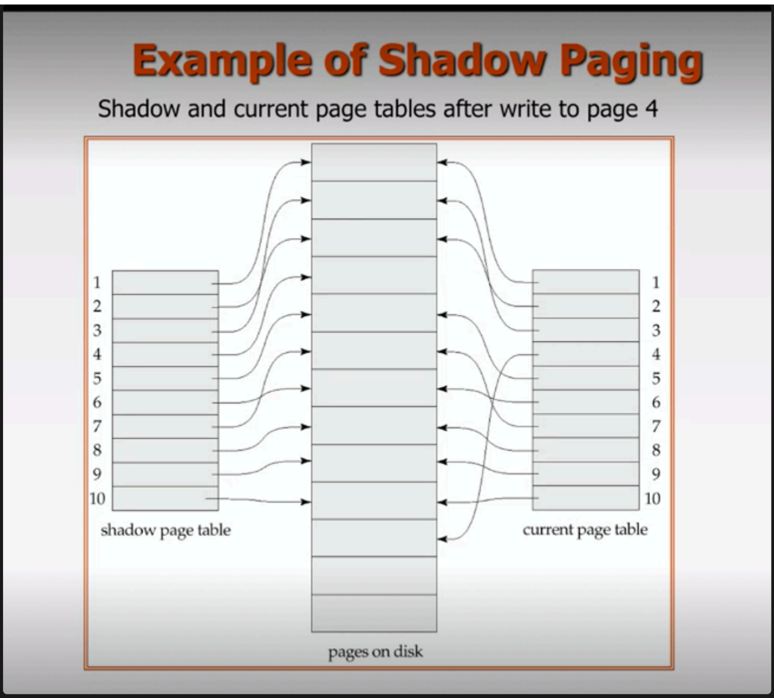
👉 log based recovery technique require recovery time (time to recovered), after system went down, it would take few minute or up to hour(for big system), what if we can afford to wait?

✅ Shadow Paging

- an alternative to log-based recovery
- allow immediate recovery
- this technique → small device (pc, handheld device), and also military system?
- minimum recovery time → no need to undo, redo



- our database will be separate into pages, and you need directory call page table
- page table have a pointer, point to pages on the disk
- this pages handle by DBMS
- when a transaction start, and the database operation work on the data → a new page table will be created



- you have shadow page table, at first it only have 1 page table.
- once, you start transaction, the all page table become shadow page table.
- and the DBMS create new page table called current page table.
- look at this diagram, page number 4, the old directory point to a place (ด้านบน), but when we modified or update page number 4, we don't modified at old place, we copy to new area, and we have pointer point to the new area.
- idea → pointer point to page in database and when any update take place, you don't modified existing one, you copy the content to a new area, and modified a new area.
- Modification take place in a new page
- System failed before commit → you restart the system, it will use shadow page, and point to the old place (switch to old data) → immediate roll back
- commit → save the current page table and discard shadow page.

> 👉 this one work for small system(with minimum update), not very good for big system with alot of update.